Vietnam National University of HCMC

International University

School of Computer Science and Engineering

Program of MIT

Course: Advanced Computer Graphics (IT516)

Lecturer: Assoc. Prof. Nguyen Van Sinh

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Course outline

- Topics:
  - Mathematics-based and Geometric modeling
  - Research topics in computer graphics (curve & surface, 3D object processing)
  - Image representation, medical image processing
  - Game application development
  - VR and AR application based on Unity 3D
- Research topics (digital heritage; eMuseum; eTourism)
- Methods, tools and skills to develop graphical applications.

The research topics & applications

Research topics:

- Background of mathematics & Geometric modeling
- Rendering and coloring techniques
- Image representation
- Medical image processing
- VR and AR application
- Visualization and simulation
- E-Museum; E-Tourism

Major applications:

- Video games, cartoons, animated films
- Visual effects
- CAD/CAM and simulation
- Medical imaging
- VR and AR applications
- Information and data visualization
- You proposed application

Lecturer information

- Lecturer:
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  - Supporter: Msc. Tran Khai Minh
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Activities

- Lectures: 09 advanced topics for teaching; 09 sessions for discussion, presentation and implementation
- Assignment & Project
- Marking schema:
  - Mid Exam (30%)
  - Presentation + Projects (30%),
  - Final Exam (40%)

Textbooks & references

Textbooks:

1) Fundamentals of Computer Graphics, 5th Edition, Steve Marschner and Peter Shirley, by A K Peters/CRC Press ISBN 9780367505035, 2021.
2) Frank Klawonn , Introduction to Computer Graphics Using Java 2D and 3D, 2nd Edition, Springer 2012 (check the latest version)

References:

1) Sumanta Guha, Computer Graphics Through OpenGL From Theory to Experiments Third Edition (AIT), CRC Press, 2019.
2) John Vince, Mathematics for Computer Graphics, 5th Edition, Springer 2017.
3) Nguyễn Văn Sinh, Trần Mạnh Hà, Nguyễn Thị Thanh Sang, Nguyễn Minh Quân, "Nền tảng toán học trong Công nghệ thông tin", Sách chuyên khảo, Nhà xuất bản Đại học Quốc gia TP.HCM, ISBN: 978-604-73-6518-0, 2018.
4) Chris Coutinho, Unity Virtual Reality Development: No C# Programming Skills Required, 2019.
5) Paul Mealy - Virtual & Augmented Reality for Dummies For Dummies 2018

Agenda

Week	Section (Handout)	Reading
1+2	Introduction to the course Mathematics for computer graphics	T1; R2
3+4	2D and 3D transformation (theory and practice)	T1,2, R1,2
5+6	Curve and Surface, theory and practice on Meshlab	T1, R2
7+8	Presentation: simplification; triangulation; hole filling on the 3D objects, both on point clouds and mesh	T2, R11,3
Review and Midterm Exam	Review and Midterm Exam	
9+10	Presentation: simplification; triangulation; hole filling on the 3D objects, both on point clouds and mesh	T1, R3
11+12	Image processing (medical image)	T1, R1, 3
13+14	VR and AR application → Unity 3D	R4,5
15+16	Digital heritage, eMuseum, eTourism	R4,5
17+18	Game industry → using different platforms	T1,2
Review and Final Exam	Review and Final Exam Revise and submit final report of project	

Tools and software for implementation

- Software:
  - Netbeans (Java 2D and 3D + JOGL)
  - VS.Net: C# on Unity 3D, Python-OpenGL
  - Meshlab: compiled version
  - Libraries: VTK, OpenGL, Point clouds, OpenCV, VCG, etc.,
- Working:
  - Read the textbooks and references
  - Revise the lectures
  - Attend class to discuss and presentation
  - Work on the exercises and assigned topics

Assignment and project in the implementation

- Work in group (2 students/group)
- Make a group and let me know
- Work at home and in the class
- Work on the same topic throughout the subject
- The list of topics are provided, proposed and suggested
- Revise, update and implementation it during your working time

Why you need to work hard?

- This subject covers several abstracts, definitions and methods of advanced topics in computer graphics
- Explore new technologies → require a lot of practices
- Important subject: knowledge in this subject is useful for your research thesis, improving jobs (engineer, master and doctoral research in computer science)

Recall the background of computer graphics

- Provides knowledge and skills to construct and handle interacting objects
  - Creation, manipulation, and storage of geometric objects (modelling) and their images (rendering)
  - Display those objects, images on the screen or hardware devices
  - Image processing
- Consists of hardware and software
- Allow developers to interact and visualize data.
- Others: GUI, Haptics, Displays (VR)...

What is the computer graphics?

- Look at following areas:
  - Hardware devices for CG
  - Rendering techniques
  - Interacting objects
  - Modelling objects & 3D Simulation
  - Visualization of data
  - VR & AR applications

Graphic hardware

- Graphic Card:
  - IS a bridge between the mainboard and display equipments (GPU)
  - Consist of a buffer memory for display frames
- Display equipments:
  - Monitor
  - Plotter (a computer printer for printing vector graphics, vector graphics devices)
- Input equipments:
  - Head, glove, positioning devices
  - Camera, 3D scanner, Lidar based sensors & laser

Input devices

Output devices

Output devices

Graphic card

- Separate graphic card
  - Have separate CPU, GPU (Graphic Processing Unit) and memory
  - Architecture and bus different with main-board
  - Accessing to buffer requires a synchronized mechanism with bus of main-board
  - Accessible by DMA (Direct Memory Access) and accelerating by AGP (Accelerated Graphics Port)
- Onboard graphic card
  - Use a part of main memory as its buffer
  - Low performance

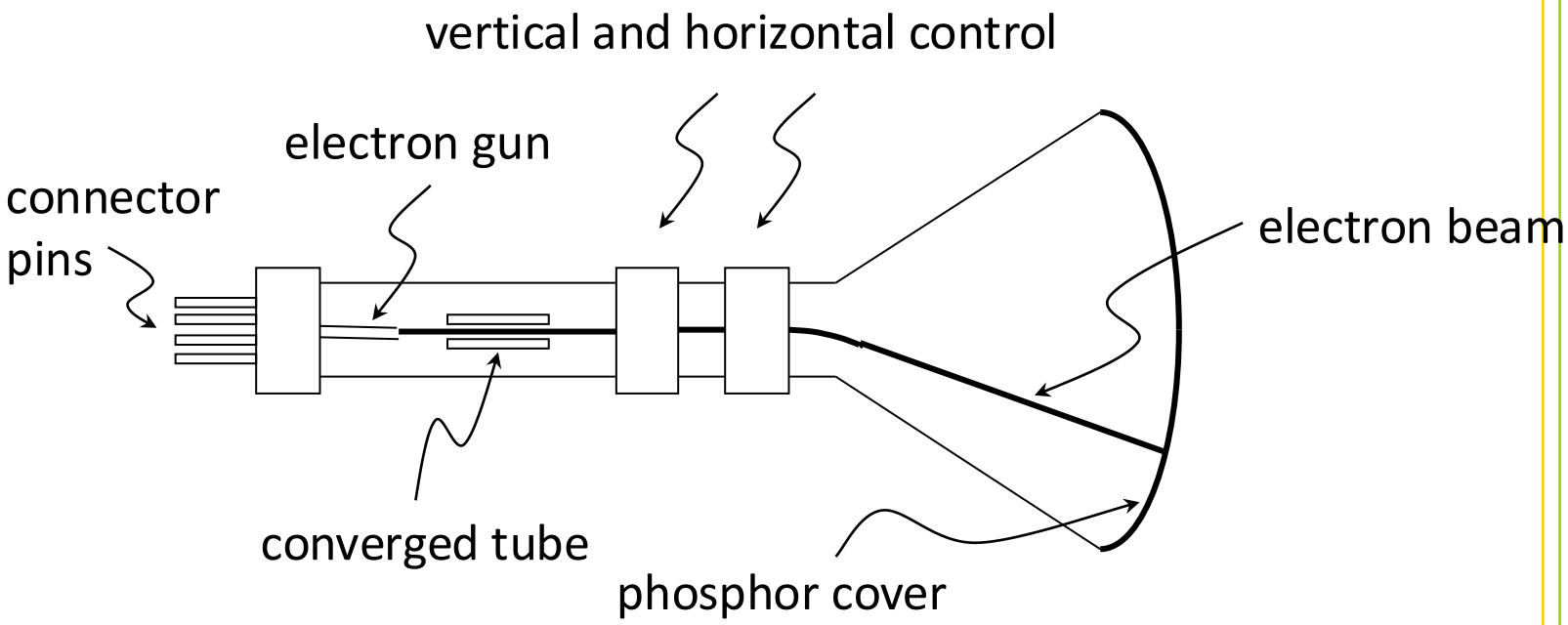
Interacting between operating system and graphic card

- Device driver: A set of routines to communicate and control graphic card
- Goal: Increase the independent with equipments
  - A common API (Application Programming Interface) to interact with various different equipments
  - Encapsulate the working with hardware, cover complicated interaction on the software

Screen resolution

- The resolution:
  - The viewable size is divided into picture elements (pixels) of horizontal (width) and vertical (height) of the screen.
- Color Depth:
  - The number of bits used to indicate the color of a single pixel.
  - The color values of each pixel = 2^{bit_color} (e.g. 8-bit = 256 colors)
- Graphics memory (Video RAM):
  - Computer memory is measured in bytes
  - Width x Height x (bit_color/8)

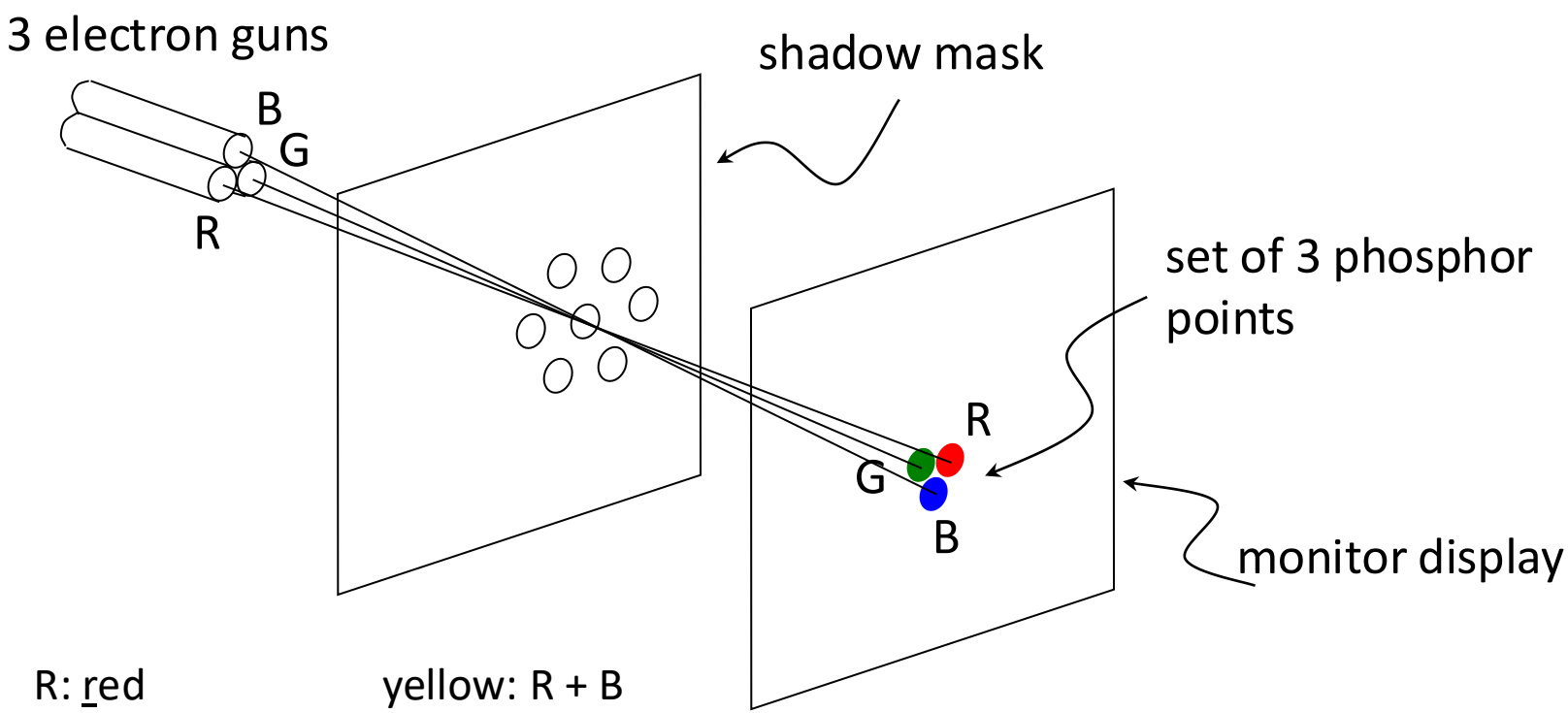
CRT



Control:

* The intensity of the electron beam
* The position the beam touches the phosphor cover

CRT Working



R: red
G: green
B: blue

yellow: R + B
white: R + G + B

LCD

- A **liquid-crystal display** (**LCD**) is defined as a flat panel display, electronic visual display, or video display
- That uses the light modulating properties of liquid crystals.
- Comparison between LCD and Plasma in the field of TV:

http://hometheater.about.com/od/lcdtvfaqs/f/lcdfaq2.htm

CRT vs LCD

- CRT
  - Advantages:
    - High resolution
    - Width and depth
    - Natural colors
    - High contrast and brightness
    - Low price
    - Wide viewing angle
    - …
  - Disadvantages:
    - Heavy
    - Higher power usage
    - Harmful to health (magnetic fields)
    - High radiation
- LCD:
  - Advantages:
    - Light
    - Low radiation or none.
    - Lower power usage
    - true perfect flat screen
    - …
  - Disadvantages:
    - High price
    - Low contrast
    - Narrow viewing angle (depending on technology employed)
    - …

Rendering

- Many think/thought graphics synonymous with rendering.
- **Rendering** is the process of generating an image from a model by means of computer programs.
- Major areas:
  - Earliest: Photo Realism
  - Recent: Non-Photorealistic Graphics (NPR)
  - Recent: Image-based Rendering (IBR)
- Ray Tracing has become practical
  - Extremely high quality images
  - Photorealism, animation, special effects
- Accurate rendering, not just pretty

Rendering realism

Photo Realism

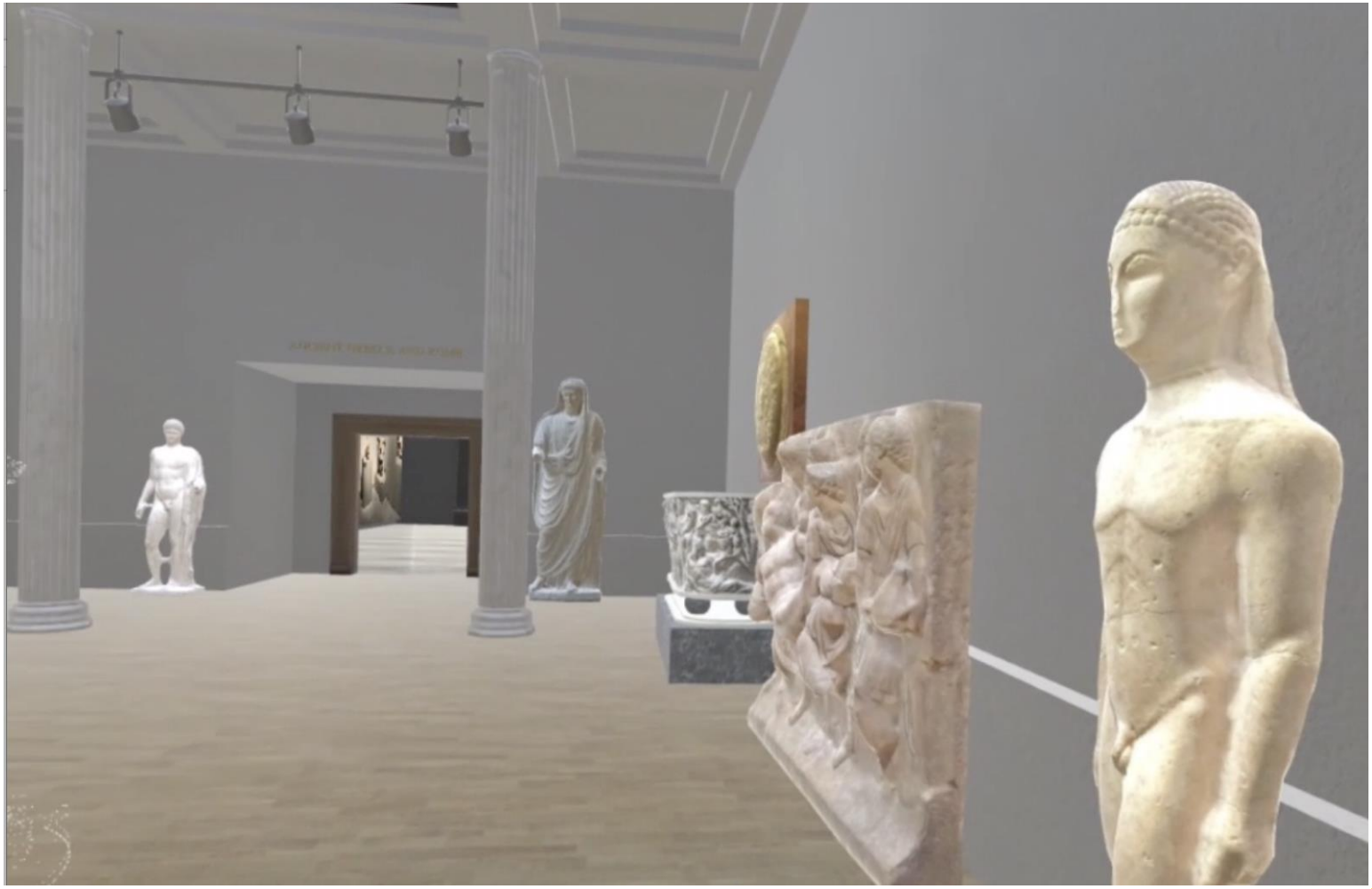
Non-Photorealistic Graphics (NPR)

Image based rendering

- Model light field
- Do not have to model geometry
- Good for complex 3D scenes
- Can leave holes where no data is available

3D scene capture

3D scene recreation viewed in VR



Interaction

- Way behind rest of graphic's spectacular advances
- Still doing WIMP:
  - Windows, icons, menus, pull-downs/pointing
- Once viewed as “soft” research
  - Turns out to be one of hardest problems

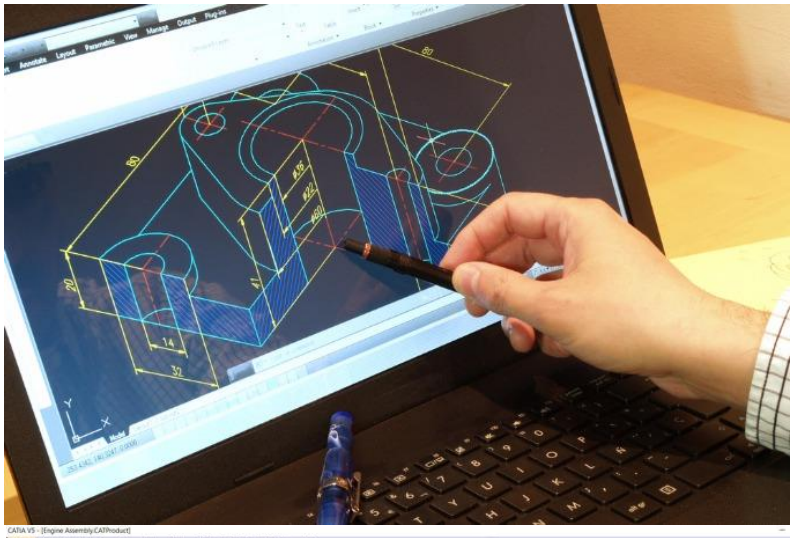
Interaction still needs…

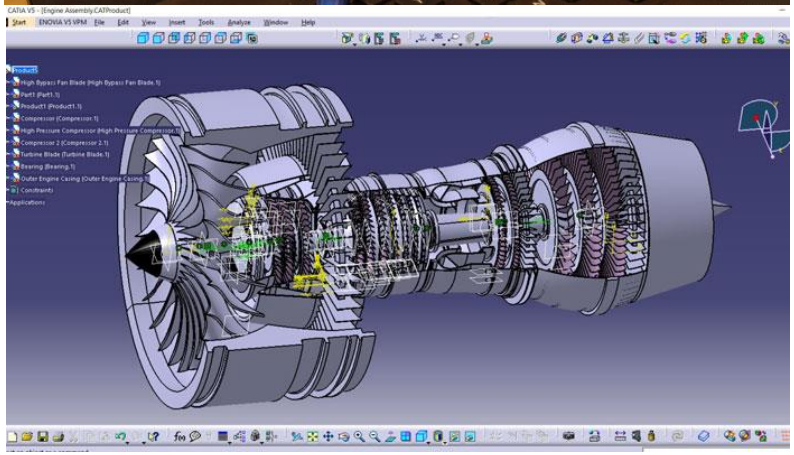
- Better input devices
- Better output devices
- Better interaction paradigms
- Better understanding of HCI
  - Bring in psychologists

Applications of computer graphics

- Computer-Aided Design for engineering and architectural systems;
- Presentation of graphical data;
- Computer Art;
- Entertainment (Games, Films);
- Education and Training;
- Visualization & simulation;
- Image Processing;
- UI & UX.

Computer Aided Design (CAD)





- Mechanical, Electronic, Architecture,...
- Drives the high end of the hardware market
- Integration of computing and display resources
- Reduced design cycles == faster systems, sooner

Movie industry

Game industry

Display scientific data

- Showing air-flows around the spaceship, car,…

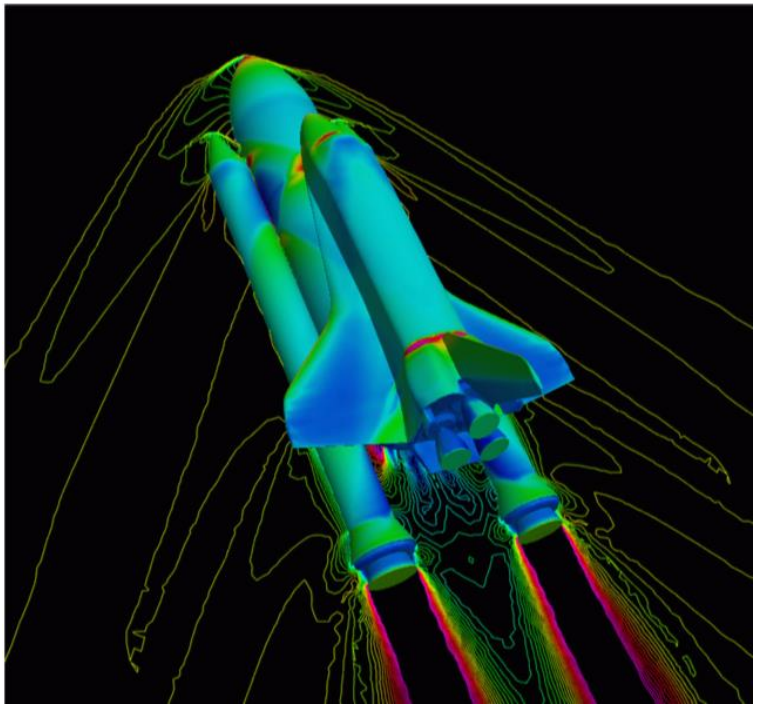
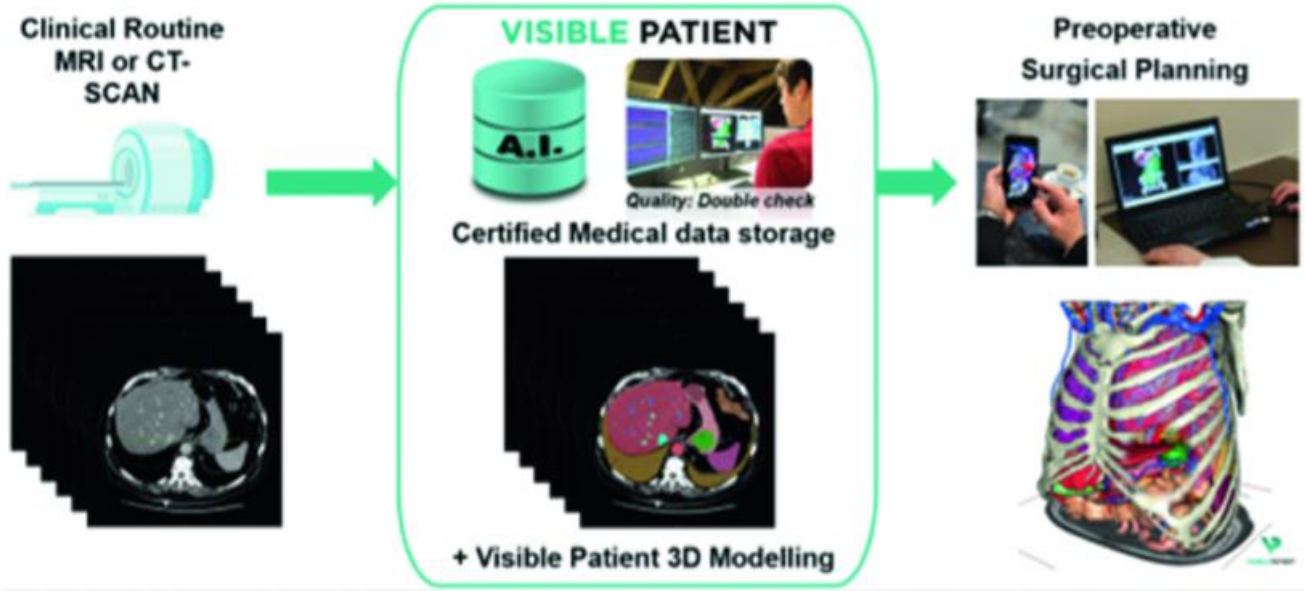


Image processing tools

- Super Paint system: Richard Shoup, Alvy Ray Smith (PARC, 1973-79)
- Photoshop.

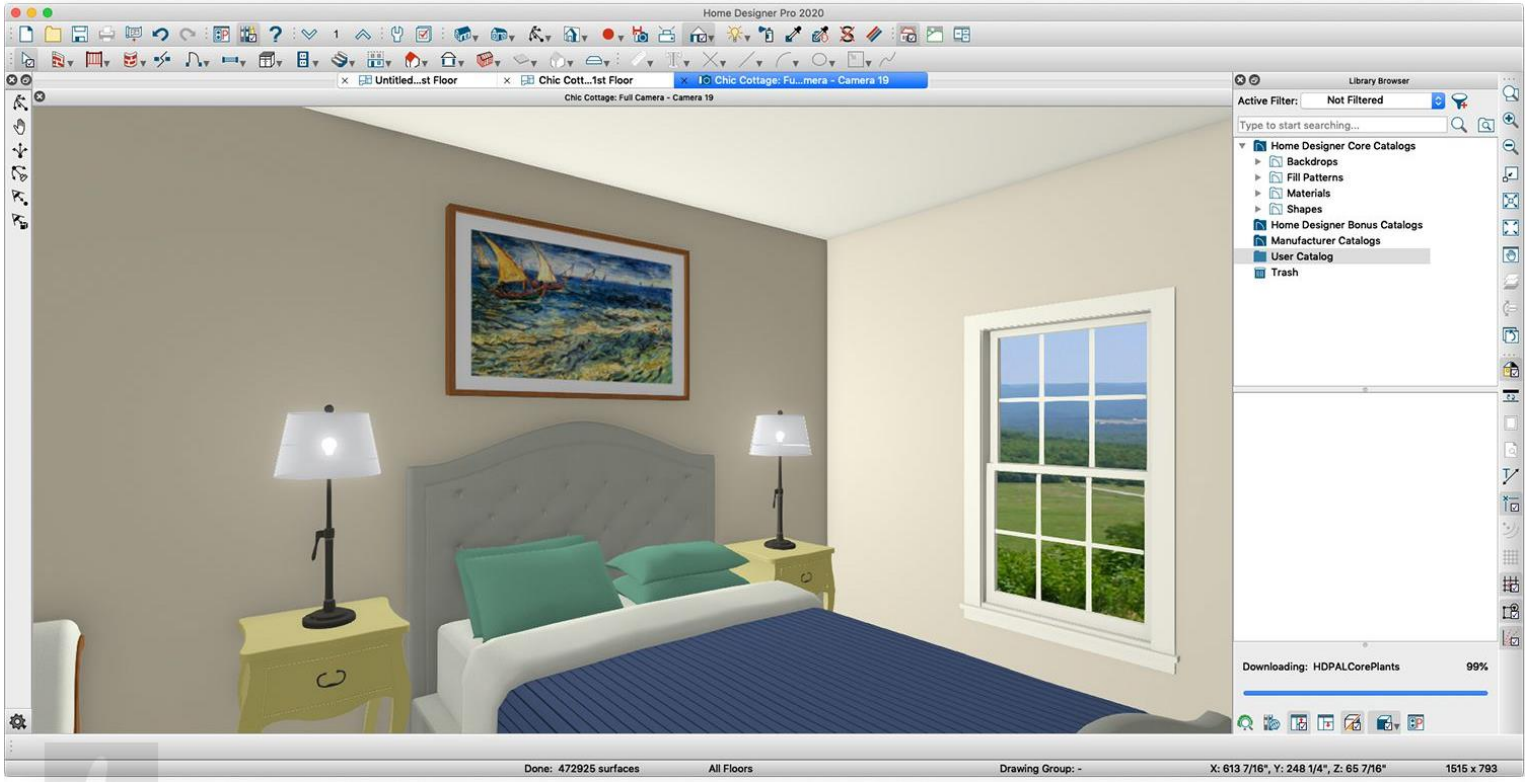
Medical imaging processing

- Tools for teaching and diagnosis
- New data representations and modalities
- Drive issues of precision and correctness
- Focus on presentation and interpretation of data
- Construction of models from acquired data



Design and simulate architectural structures

- Home Designer & Architecture



GIS (Geographic Information System)

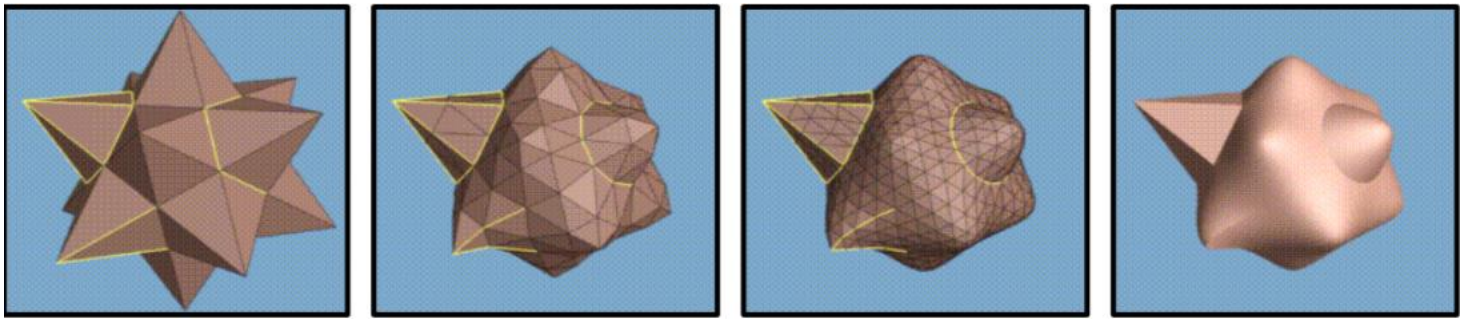
Google Map **Google Earth** **Uber** **Grab**

Geometric modeling

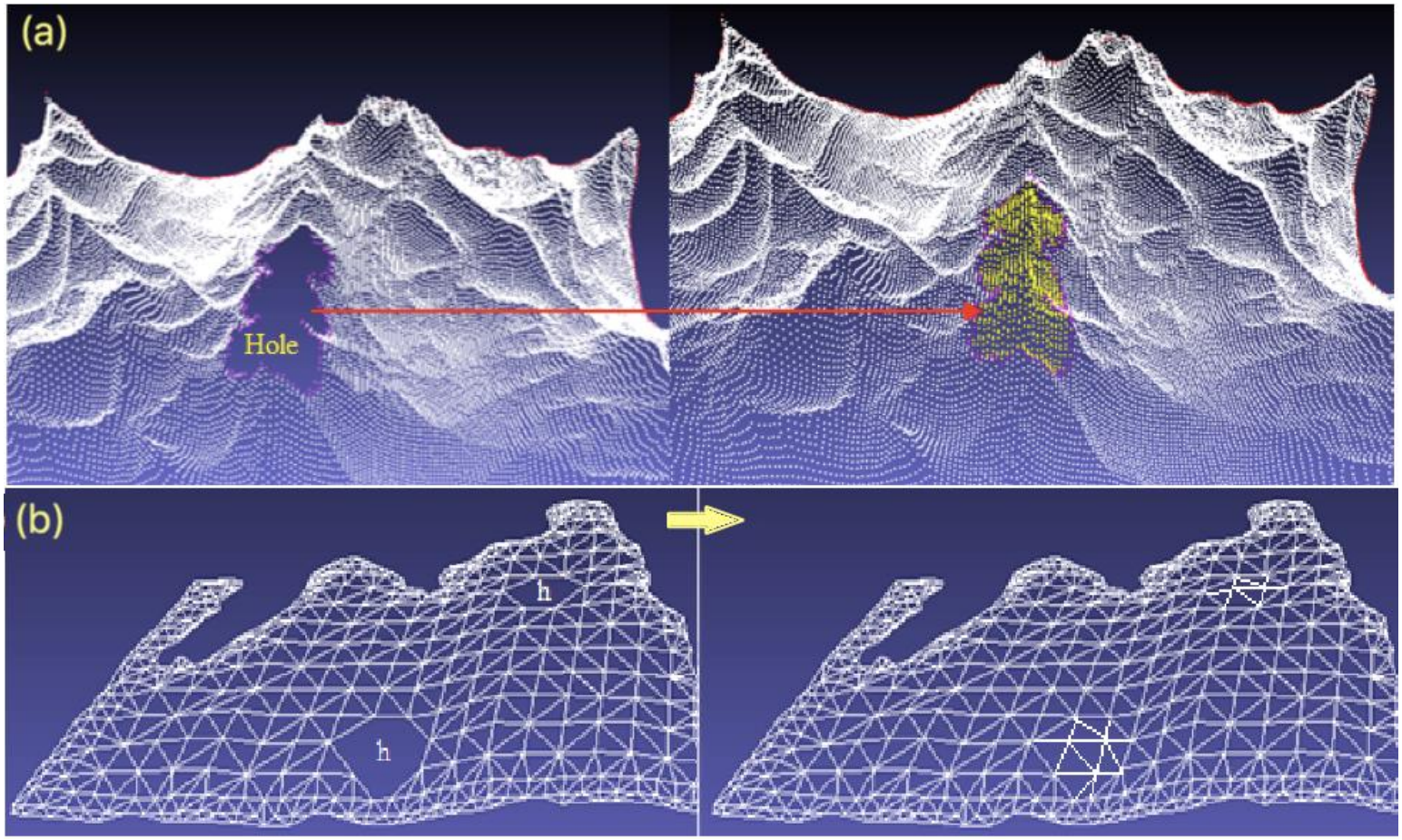
- A field of research based on applied mathematics and computational geometry.
- Modeling is hard:
  - Mathematical challenge
  - Computational challenge
  - Interaction challenge
  - Display challenge (want 3D)
  - Domain knowledge, constraints

Many model representations

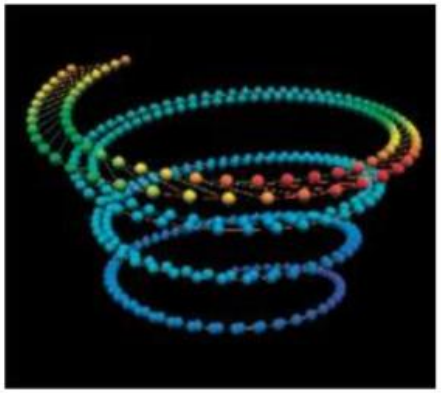
- Bezier, B-spline, box splines, simple splines, polyhedral splines, quadrics, super-quadrics, implicit, parametric, subdivision, fractal, level sets, etc.

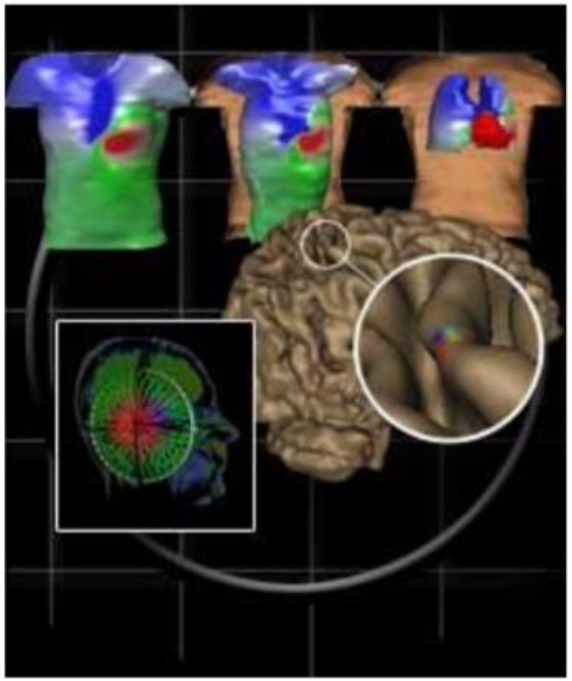


Reconstructing the 3D models



Scientific visualization





Computer graphics libraries

- OpenGL (Open Graphics Library): http://www.opengl.org
- VCG (Visualization and Computer Graphics Library): http://vcg.isti.cnr.it
- VTK (Visualization Toolkit): http://www.vtk.org
- CGAL (Computational Geometry Algorithms Library): http://www.cgal.org
- PCL (Point Cloud Library): http://pointclouds.org
- https://unity.com
- https://arvr.google.com